Mixture of Sub-manifolds Non-negative Matrix Factorization (MoS-NMF)

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Abstract

1 Introduction

Considering Poisson-NMF likelihood:

$$Y_{nm} \sim \text{Poi}\left(\sum_{k=1}^{K} L_{nk}F_{km}\right),$$

for count data $(Y_{nm})_{n,m=1}^{N,M}$, where Y_{nm} counts how many times item m appears in data/cell/document n. $L \in \mathbb{R}_+^{N\times K}$ capture the loadings/weights and $F \in \mathbb{R}_+^{K\times M}$ are K factors/parts. L is the observation's specific score.

Modeling goal. We want to model that each Y_n is a combination of not all K parts, but only a few of them. For example, even if we fit $K = 50$ topics to the copura, it makes sense to assume that each document can only discuss a maximum of 3-5 topics. Similarly, in music, each song can only be mixed of at most 2-3 genres. So we want to model each loading $L_n \in \mathbb{R}_+^K$ to have at most D non-zero effects. There are many ways to choose a combination of D topics from K topics, and we call each such way a "motif". Hence, we have S sub-manifolds, each of which has a dimension of at most D in $\mathbb{R}_+^K$, which is controlled by parameter $\square_s := (\gamma_s, \alpha_s^0, \lambda_s^0)$ for $\gamma_s = (\gamma_{sd})_{d=1}^D \in [K]^D$ indicates which D topics are chosen and $(\alpha_{sd}^0, \lambda_{sd}^0)_{s,d=1}^{S,D}$ controls the randomness of the effect size. In other word, given the sub-manifold Z_n of the loading L_n, we have

$$(L_n|Z_n = s) \sim \sum_{d=1}^{D} e_{\gamma_{sd}}\beta_{nd}, \quad \beta_{nd} \sim \text{Gam}(\alpha_{sd}^0, \lambda_{sd}^0), \tag{1}$$

where e_k is the vector of 0 except a single 1 at k-th element. We call each of these distributions $\text{Gam}(\square_s)$. Thus each L_n is modeled as a mixture of them:

$$L_n \overset{iid}{\sim} \sum_{s=1}^{S} \pi_s \text{Gam}(\square_s). \tag{2}$$

We finish the model by putting uniform prior on each γ_s:

$$\gamma_{sd} \overset{iid}{\sim} \text{Unif}([K]), \quad s \in [S], d \in [D]. \tag{3}$$

The observation is

$$Y_{nm}|L_n, F \overset{ind.}{\sim} \text{Poi}(\sum_{k=1}^{K} L_{nk}F_{km}), \quad n \in [N], m \in [M]. \tag{4}$$

We will make inference of $(\pi_s)_{s=1}^S, (\alpha_{sd}^0, \lambda_{sd}^0)_{s,d=1}^{S,D}$ and F by mean of Maximum Likelihood, and make inference of indicator γ and effect size $(\beta_{nd})_{N,D}$ by its posterior (given the MLE of other parameters). So this is an Empirical Bayes procedure. The reason why we want to use the Bayesian formula for the indicator γ is that we think it can quantify the uncertainty if there are some very similar F_k's (detect over-fitting). Let's see if it works as intended.

The whole model. Hence, the whole model is as follows:

$$\begin{cases} \gamma_{sd} \overset{iid}{\sim} \mathrm{Unif}([K]), & s \in [S], d \in [D] \\ Z_n|\pi \overset{iid}{\sim} \mathrm{Cat}(\pi) \in [S], & n \in [N] \\ \beta_{nd}|(\alpha^0, \lambda^0, Z_n = s) \overset{ind.}{\sim} \mathrm{Gam}(\alpha_{sd}^0, \lambda_{sd}^0), & n \in [N], s \in [S], d \in [D] \\ Y_{nm}|(\gamma, \beta, Z_n = s) \overset{ind.}{\sim} \mathrm{Poi}(\sum_{d=1}^D \beta_{nd} F_{\gamma_{sd} m}), & n \in [N], m \in [M], \end{cases} \tag{5}$$

with latent variables $(Z_n)_{n=1}^N, (\gamma_{sd})_{s,d=1}^{S,D}, (\beta_{nd})_{n,d=1}^{N,D}$ and parameters $(\pi_s)_{s=1}^S, (\alpha_{sd}^0)_{s,d=1}^{S,D}, (\lambda_{sd}^0)_{s,d=1}^{S,D}$, and $F \in \mathbb{R}_+^{K\times M}$. For the identifiability reason, we will assume that $\sum_m F_{km} = 1$ for all $k \in [K]$.

2 Preliminaries

2.1 Poisson SuSiE

MoS-NMF is a complex hierarchical model that consists of (1) Poisson-SuSiE components and (2) mixture modeling. In this section, we present a backbone SuSiE-Poisson regression model for one observation and its inference algorithm, which helps build intuition and can later be incorporated into the inference of the MoS-NMF model. Given an observation $y = (y_1, \ldots, y_m) \in \mathbb{N}^M$ and factors $F \in \mathbb{R}_+^{K\times M}$, consider the sparse regression model

$$y \sim \mathrm{Poi}(lF),$$

where $l \in \mathbb{R}_+^K$, and the Poisson distribution is independent across M dimensions, i.e.,

$$y_m \overset{ind.}{\sim} \mathrm{Poi}(\sum_{k=1}^K l_k F_{km}), \quad m \in [M].$$

We want to model the sparsity of $l \in \mathbb{R}^K$ using the Sum of Single Effect model with at most D non-zero effect:

$$l = \sum_{d=1}^D l^{(d)} = \sum_{d=1}^D e_{\gamma_d} \beta_d,$$

where $\gamma_d \sim \mathrm{Unif}([K])$ (sparsity index indicator) and $\beta_d \sim \mathrm{Gamma}(\alpha_d^0, \lambda_d^0)$ a priori (Gamma distribution is parameterized using shape-rate parameter here). We want to make inference of the posterior $(\gamma_d, \beta_d)_{d=1}^D|y$.

Poisson Single Effect Regression. When $D = 1$, we can explicitly calculate the posterior

$$
\begin{aligned}
p(y|\gamma_1 = k) &= \int p(x|\gamma_1 = k, \beta_1)\mathrm{Gamma}(\beta_1|\alpha_1^0, \lambda_1^0)d\beta_1 \\
&= \int \prod_{m=1}^{M} \mathrm{Poi}(y_m|\beta_1 F_{km})\mathrm{Gamma}(\beta_1|\alpha_1^0, \lambda_1^0)d\beta_1 \\
&= \int_{\beta_1} \frac{e^{-\beta_1 \sum_m F_{km}} \prod_m (\beta_1 F_{km})^{y_m}}{\prod_m y_m!} \frac{(\lambda_1^0)^{\alpha_1^0}}{\Gamma(\alpha_1^0)} e^{-\beta_1 \lambda_1^0} \beta_1^{\alpha_1^0 - 1} d\beta_1 \\
&= \frac{\prod_{m=1}^{M} F_{km}^{y_m}}{\prod_{m=1}^{M} y_m!} \frac{(\lambda_1^0)^{\alpha_1^0}}{\Gamma(\alpha_1^0)} \frac{\Gamma(\sum_m y_m + \alpha_1^0)}{(\sum_m F_{km} + \lambda_1^0)^{\sum_m y_m + \alpha_1^0}}.
\end{aligned}
$$

Hence,

$$
p(\gamma_1 = k|y) \propto \frac{\prod_{m=1}^{M} F_{km}^{y_m}}{(\sum_m F_{km} + \lambda_1^0)^{\sum_m y_m + \alpha_1^0}}, \quad k \in [K]. \tag{6}
$$

Moreover, by the Poisson-Gamma conjugacy, it is straightforward that:

$$
p(\beta_1|y, \gamma_1 = k) \sim \mathrm{Gam}\left(\sum_m y_m + \alpha_1^0, \sum_m F_{km} + \lambda_1^0\right), \quad k \in [K]. \tag{7}
$$

Hence, the posterior distribution $p(\gamma_1, \beta_1|x)$ can be summarized by $3K$ non-negative number. (This point can be simplified by noticing the posterior of β_1.) For the sake of future argument, we also note that this exact posterior distribution has the variational characterization that

$$
p(\gamma_1, \beta_1|y) = \underset{q(\gamma_1,\beta_1)}{\arg\max}\, \mathbb{E}_q \left[\sum_m \log \mathrm{Poi}(y_m|\beta_1 F_{\gamma_1 m})\right] - \mathrm{KL}(q(\gamma_1, \beta_1)\|p(\gamma_1, \beta_1)), \tag{8}
$$

where $\mathrm{Poi}(y_m|\beta_1 F_{\gamma_1 m}) = y_m \log(\beta_1) + \log(F_{\gamma_1 m}) - F_{\gamma_1 m}\beta_1 - \log(y_m!)$

Poisson SuSiE. The posterior of $(\gamma_d, \beta_d)_{d=1}^{D}$ does not have a simple closed-form expression, so we use the same Variational approximation as in Gaussian SuSiE Wang et al. (2020):

$$
p(\gamma, \beta|y) \approx \prod_{d=1}^{D} q_d(\gamma_d, \beta_d) = \prod_{d=1}^{D} (q(\gamma_d)q(\beta_d|\gamma_d)), \tag{9}
$$

and find the optimal variational distribution by maximizing the ELBO:

$$
\mathbb{E}_{\prod_d q_d}\left[\sum_m \log \mathrm{Poi}\Big(y_m \mid \sum_{d=1}^{D} \beta_d F_{\gamma_d m}\Big)\right] - \sum_{d=1}^{D} \mathrm{KL}(q_d(\gamma_d, \beta_d)\|p(\gamma_d, \beta_d)).
$$

Note that in the log-likelihood:

$$
\log \mathrm{Poi}\Big(y_m \mid \sum_{d=1}^{D} \beta_d F_{\gamma_d m}\Big) = y_m \log\left(\sum_{d=1}^{D} \beta_d F_{\gamma_d m}\right) - \sum_{d=1}^{D} \beta_d F_{\gamma_d m} - \log(y_m!),
$$

the third term is a constant, the expectation of the second term can be calculated simply as

$$\mathbb{E}_q \sum_{d=1}^{D} \beta_d F_{\gamma_d m} = \sum_{d=1}^{D} \sum_{k=1}^{K} p(\gamma_d = k) \mathbb{E}[\beta_d | \gamma_d = k] F_{km}, \tag{10}$$

while the first term does not have a simple closed-form expectation. So that we utilize a well-known trick in latent variable model to introduce new auxiliary variables $(\xi_{md})_{m,d=1}^{M,D}$ and use the lower bound:

$$\log\left(\sum_{d=1}^{D} \beta_d F_{\gamma_d m}\right) \geq \sum_{d=1}^{D} \xi_{md} \left(\log\left(\frac{\beta_d F_{\gamma_d m}}{\xi_{md}}\right)\right), \tag{11}$$

where $\sum_d \xi_{md} = 1$ for all $m \in [M]$. So that we are maximizing the following Evidence Lower Bound:

$$\begin{aligned} \mathcal{E}(q) = & \sum_{d=1}^{D} \mathbb{E}_{q_d(\gamma_d, \beta_d)} \left[\sum_m \left(\xi_{md} y_m (\log(\beta_d) + \log(F_{\gamma_d m})) - \beta_d F_{\gamma_d m}\right)\right] \\ & - \sum_{d=1}^{D} \mathrm{KL}(q(\gamma_d, \beta_d) \| p(\gamma_d, \beta_d)) - \sum_{m,d} y_m \xi_{md} \log(\xi_{md}). \end{aligned} \tag{12}$$

Algorithm 1 SuSiE-Poisson algorithm

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for Outer loop **do**
  for d from 1 to D **do**
    Update auxiliary variable $\xi_{md} \leftarrow \text{Norm-Exp}(\mathbb{E}_{q(\gamma_d,\beta_d)}(\log \beta_d + \log(F_{\gamma_d m})))$ $\forall m \in [M], d \in [D]$. ▷ (Gauss-Seidel style update)
    Update $\alpha_{dk} \leftarrow \sum_m \xi_{md} y_m + \alpha_d^0$ ▷ (Variational parameters)
    Update $\lambda_{dk} \leftarrow \sum_m F_{km} + \lambda_d^0$ ▷ (Variational parameters)
    Assign $q(\beta_d | \gamma_d = k) \equiv \mathrm{Gam}(\alpha_{dk}, \lambda_{dk})$ ▷ (Variational distributions)
    Calculate $\text{logprob}_{dk} = \sum_m \xi_{md} y_m \log(F_{km}) - (\sum_m \xi_{md} y_m + \alpha_d^0) \log(\sum_m F_{km} + \lambda_d^0)$
    Update $q(\gamma_d) = (\bar{\gamma}_{dk})_{k=1}^K \leftarrow \text{Norm-Exp}(\text{logprob}_{d1}, \ldots, \text{logprob}_{dK})$ ▷ (PIP)
    Update $(\alpha_d^0, \lambda_d^0)$ ▷ (Optional; VEB)
  end for
end for

ELBO. Let $q(\gamma_d = k) \equiv \bar{\gamma}_{dk}$ and $q(\beta_d | \gamma_d = k) \equiv \mathrm{Gam}(\alpha_{dk}, \lambda_{dk})$, where

$$\alpha_{dk} = \sum_m \xi_{md} y_m + \alpha_d^0, \quad \lambda_{dk} = \sum_m F_{km} + \lambda_d^0.$$

We have

$$\mathbb{E}_{q(\beta_d|\gamma_d=k)}[\log(\beta_d)] = \psi(\alpha_{dk}) - \log(\lambda_{dk}), \quad \mathbb{E}_{q(\beta_d,\gamma_d)}[\log(\beta_d)] = \sum_k \bar{\gamma}_{dk}[\psi(\alpha_{dk}) - \log(\lambda_{dk})], \tag{13}$$

and

$$\begin{aligned}\mathrm{KL}(q_d \| p_d) &= \sum_{k=1}^{K} \bar{\gamma}_{dk}(\log(\bar{\gamma}_{dk}) - \log(\tfrac{1}{K})) \\ &+ \sum_{k=1}^{K} \bar{\gamma}_{dk}\left[(\alpha_{dk} - \alpha_d^0)\psi(\alpha_{dk}) + (\log(\lambda_{dk}) - \log(\lambda_d^0))\alpha_d^0 - \log\frac{\Gamma(\alpha_{dk})}{\Gamma(\alpha_d^0)} - (\lambda_{dk} - \lambda_d^0)\frac{\alpha_{dk}}{\lambda_{dk}}\right].\end{aligned} \tag{14}$$

Remark 2.1. Our experiments show that updating ξ every d (Gauss-Seidel style) helps the algorithm escape local optima much more frequently than updating ξ only once per outer loop (Jacobi style).

2.2 Mixture modeling

3 Model fitting in MoS-NMF

We approximate the posterior as:

$$p(Z, \gamma, \beta | X) \approx q(Z, \gamma, \beta) = \prod_{n=1}^{N} q(Z_n) \prod_{s,d=1}^{S,D} q(\gamma_{sd}) \prod_{n,d} q(\beta_{nd} | Z_n, \gamma_{Z_n d}).$$

Should I factor β, or the variational lower bound implies it? Note that both Z and γ are discrete. Z_n takes value in $[S]$, and γ in $[K]$. Write

$$\tilde{z}_{ns} := q(Z_n = s), \quad \tilde{\gamma}_{sdk} := q(\gamma_{sd} = k).$$

Also notice that the likelihood can be decomposed as:

$$\log p(X | Z, \gamma, \beta) = \sum_{n=1}^{N} \log p(Y_n | Z_n, \gamma, \beta_n), \quad \log p(\beta | Z, \alpha^0, \lambda^0) = \sum_{n=1}^{N} p(\beta_n | Z_n, \alpha^0, \lambda^0),$$

where

$$\log p(Y_n | Z_n = s, \gamma, \beta_n) = \sum_m \log \mathrm{Poi}(Y_{nm} | \sum_{d=1}^{D} \beta_{nd} F_{\gamma_{sd} m}).$$

and

$$\log p(\beta_n | Z_n = s, \alpha^0, \lambda^0) = \sum_{d=1}^{D} \log \mathrm{Gam}(\beta_{nd} | \alpha_{sd}^0, \lambda_{sd}^0).$$

Using variational lower bound:

$$
\begin{aligned}
p(X) \geq \mathbb{E}_{q(Z,\gamma,\beta)}\big\{ &\log p(X|Z,\gamma,\beta) + \log p(\beta|\alpha^0,\lambda^0,Z) + \log p(Z|\pi) + \log p(\gamma) \\
& - \log q(Z) - \log q(\gamma) - \log q(\beta|Z,\gamma)\big\} \\
= \sum_{n=1}^{N}\sum_{s=1}^{S} \tilde{z}_{ns}\mathbb{E}_{q(\gamma)q(\beta|Z,\gamma)}\big\{ &\log p(Y_n|Z_n = s,\gamma,\beta) + \log p(\beta_n|Z_n = s,\alpha^0,\lambda^0) \\
& + \log p(Z_n = s|\pi) + \log p(\gamma) - \log q(Z_n = s) - \log q(\gamma) - \log q(\beta_n|Z_n = s,\gamma)\big\} \\
= \sum_{n=1}^{N}\sum_{s=1}^{S} \tilde{z}_{ns}\mathbb{E}_{q(\gamma)q(\beta|Z,\gamma)}\Bigg\{ &\sum_{m=1}^{M} \log \mathrm{Poi}\big(Y_{nm}|\sum_d \beta_{nd}F_{\gamma_{sd}m}\big) + \sum_d \log \mathrm{Gam}(\beta_{nd}|\alpha^0_{sd},\lambda^0_{sd}) \\
& + \log p(\gamma) - \log q(\gamma) - \log q(\beta_n|Z_n = s,\gamma)\Bigg\} + \sum_{n=1}^{N}\sum_{s=1}^{S} \tilde{z}_{ns}\big(\log(\pi_s) - \tilde{z}_{ns}\big)
\end{aligned}
$$

To efficiently estimate the expectation of the log-likelihood with respect to $q(\gamma)$, we further use a lower bound with auxiliary variables $(\xi_{nmd})_{n,m,d}^{N,M,D}$:

$$
\begin{aligned}
&\log \mathrm{Poi}\big(Y_{nm}|\sum_d \beta_{nd}F_{\gamma_{sd}m}\big) = -\sum_d \beta_{nd}F_{\gamma_{sd}m} + Y_{nm}\log\left(\sum_d \beta_{nd}F_{\gamma_{sd}m}\right) - \log(Y_{nm}!) \\
&\geq \sum_d \left(-\beta_{nd}F_{\gamma_{sd}m} + Y_{nm}\xi_{nmd}\log\left(\frac{\beta_{nd}F_{\gamma_{sd}m}}{\xi_{nmd}}\right)\right) - \log(Y_{nm}!)
\end{aligned} \tag{15}
$$

Ignoring the constant terms $\log(Y_{nm}!)$ and the terms $\log p(\gamma)$ (uniform distributions), since now the log-likelihood's lower bound can be written as a sum over d, we can now write the lower bound as:

$$
\begin{aligned}
p(X) \geq \sum_{n,s=1}^{N,S} \tilde{z}_{ns}\mathbb{E}_{q(\gamma)q(\beta|Z,\gamma)}\Bigg\{ &\sum_{d,m=1}^{D,M}\left(-\beta_{nd}F_{\gamma_{sd}m} + Y_{nm}\xi_{nmd}\log\left(\frac{\beta_{nd}F_{\gamma_{sd}m}}{\xi_{nmd}}\right)\right) \\
& + \sum_d \log \mathrm{Gam}(\beta_{nd}|\alpha^0_{sd},\lambda^0_{sd}) + \log p(\gamma) - \log q(\gamma) - \log q(\beta_n|Z_n = s,\gamma)\Bigg\} \\
& + \sum_{n=1}^{N}\sum_{s=1}^{S} \tilde{z}_{ns}\big(\log(\pi_s) - \tilde{z}_{ns}\big) \\
\geq \sum_{n,s,d,k=1}^{N,S,D,K} \tilde{z}_{ns}\tilde{\gamma}_{sdk}\mathbb{E}_{q(\beta|Z,\gamma)}\Bigg\{ &\sum_{m=1}^{M}\left(-\beta_{nd}F_{km} + Y_{nm}\xi_{nmd}\log\left(\frac{\beta_{nd}F_{km}}{\xi_{nmd}}\right)\right)
\end{aligned}
$$

Lemma 3.1. *For summary statistics $(T^1_{nd}, T^{\log}_{nd}, T^0_{nd})$ we have*

$$
\begin{aligned}
&\max_q \mathbb{E}_{q(\beta_{nd})}\left[-T^1_{nd}\beta_{nd} + (T^{\log}_{nd} - 1)\log(\beta_{nd}) + T^0_{nd} - \log q(\beta_{nd})\right] \\
&= T^0_{nd} - \log(T^1_{nd})T^{\log}_{nd} + \log\Gamma(T^{\log}_{nd}).
\end{aligned}
$$

Proof.[Proof of Lemma 3.1] By means of the variational principle,

$$
q(\beta_{nd}) \propto \beta_{nd}^{(T^{\log}_{nd}-1)} e^{-\beta_{nd}T^1_{nd}},
$$

which implies $q(\beta_{nd}) = \mathrm{Gam}(T_{nd}^{\log}, T_{nd}^{1})$. Hence,

$$\log q(\beta_{nd}) = (T_{nd}^{\log} - 1)\log(\beta_{nd}) - T_{nd}^{1}\beta_{nd} + \log(T_{nd}^{1})T_{nd}^{\log} - \log\Gamma(T_{nd}^{\log}). \tag{16}$$

Plug this in, and we have the desire equation. □

Thoughts

To fix the identifiability issue of the scales, we can either choose to

1. Fix the scale of F, i.e., force $\sum_m F_{km} = 1$ for all k and learn λ_{sd}^{0} for all s, d
2. Let F be free and fix $\lambda_{sd}^{0} = 1$ for all s, d.

In the case that the modeling assumption of Gamma distribution does not hold, I think the first strategy leads to over-specifying S, while the second strategy leads to over-specifying K. I like the first one better.

References

Wang, G., Sarkar, A., Carbonetto, P., and Stephens, M. (2020). A simple new approach to variable selection in regression, with application to genetic fine-mapping. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 82(5):1273–1300. (Cited on page 3.)